ME 165
Basic Mechanical Engineering

Lecture 11

Kinematics of Rigid Bodies

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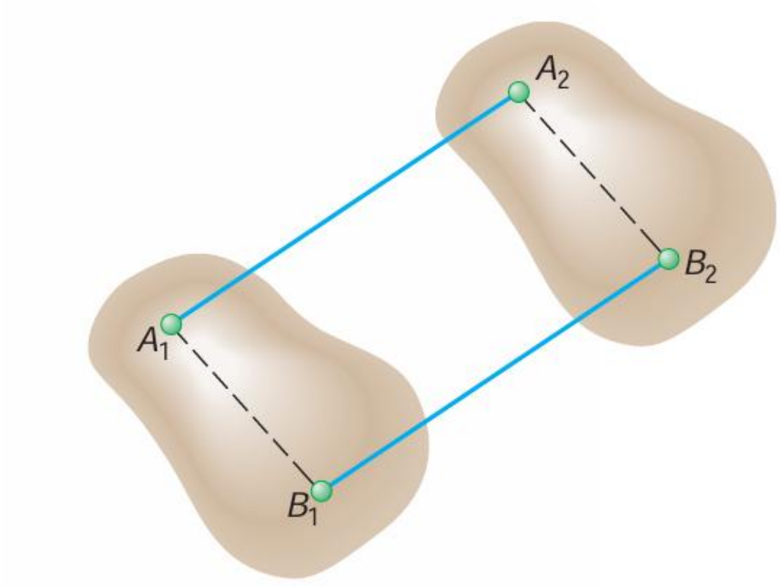
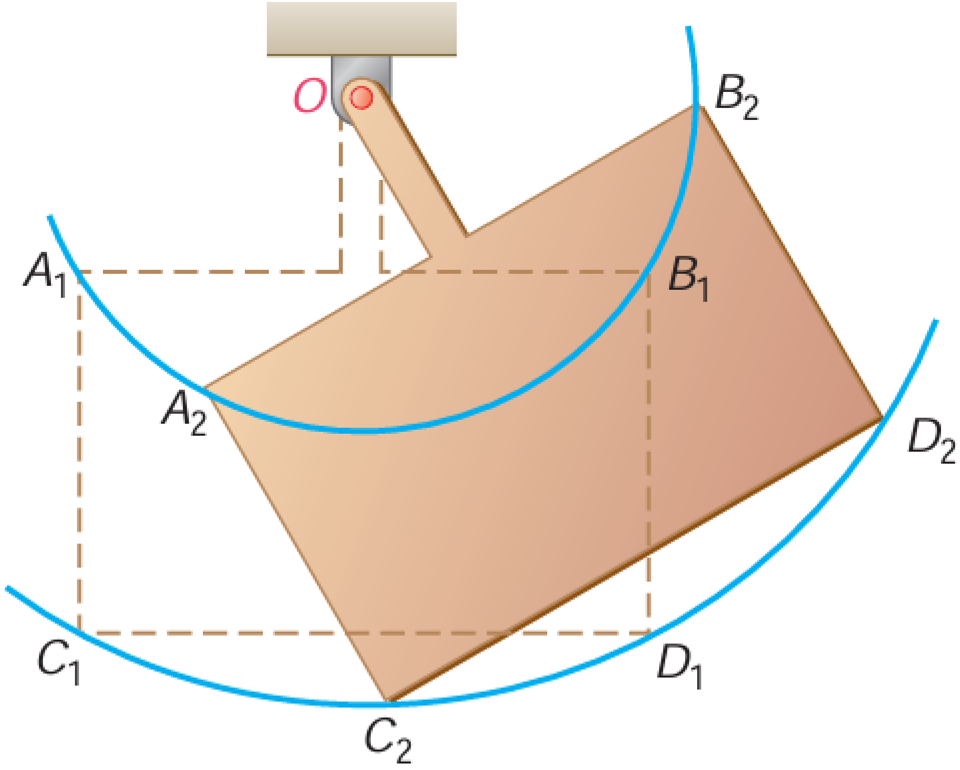
Kinematics of Rigid Bodies

Translation

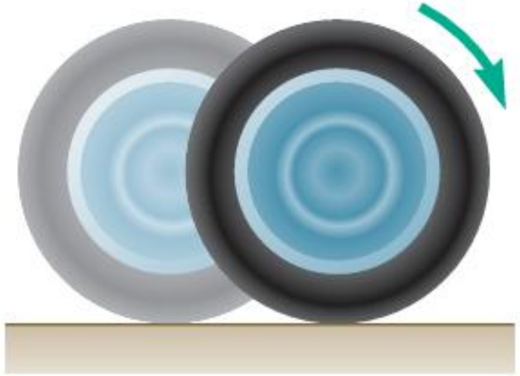


Rotation About a Fixed Axis

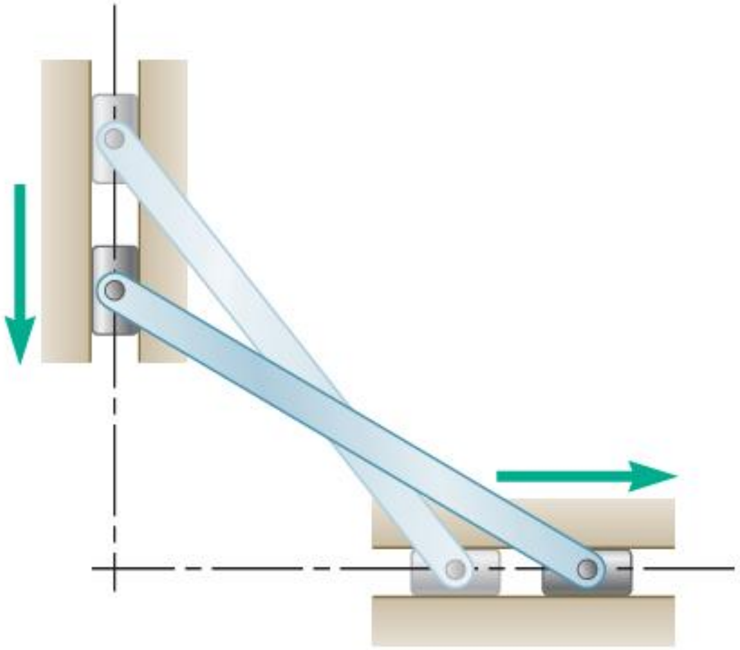


Kinematics of Rigid Bodies

General Plane Motion

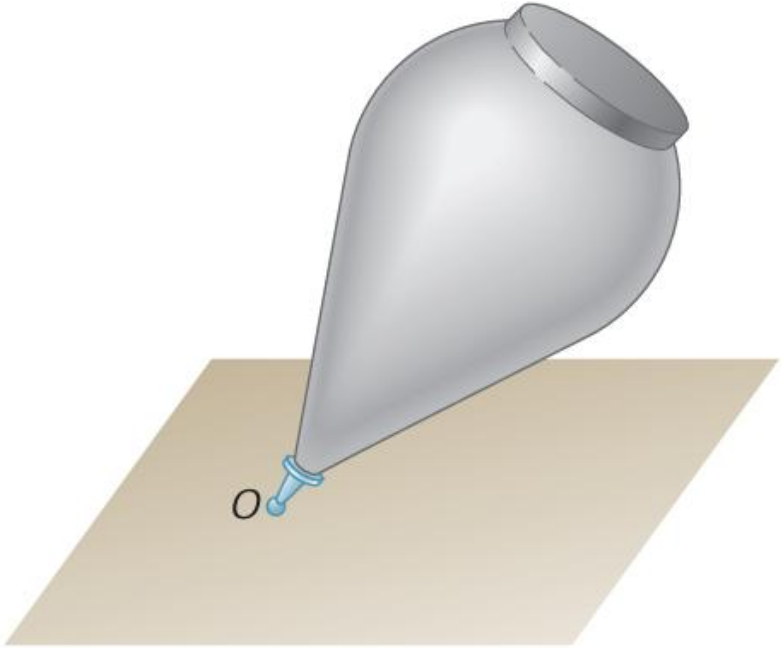


(a) Rolling wheel



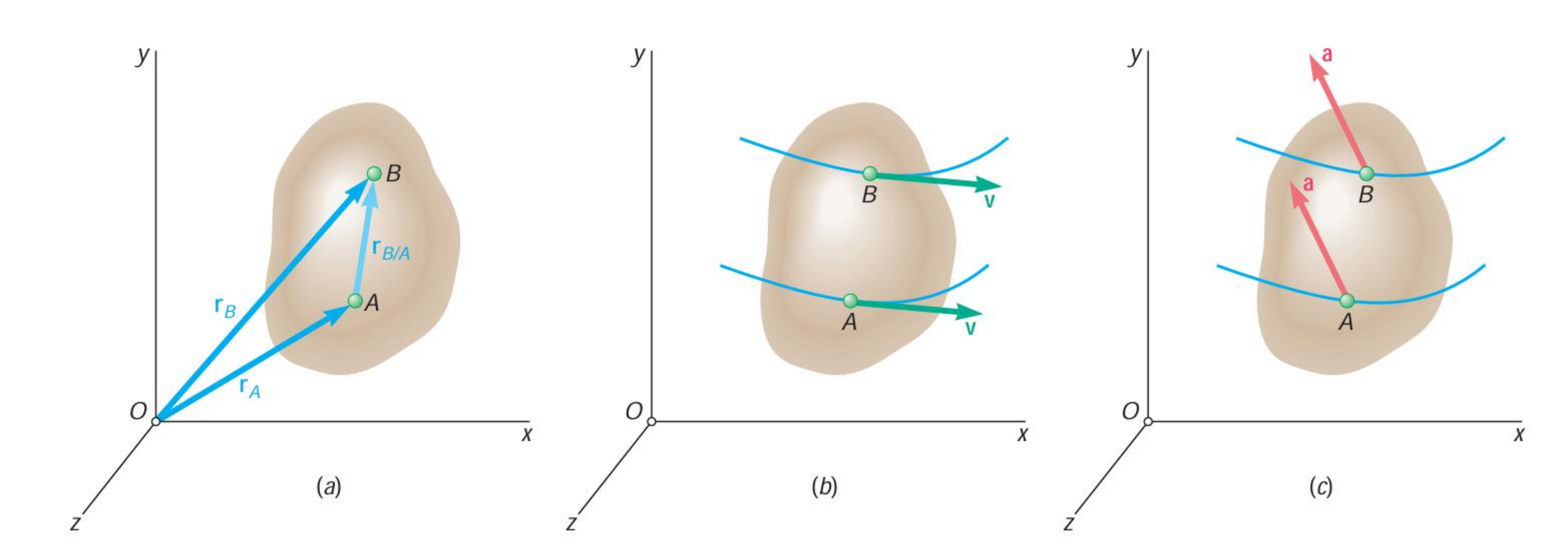
(b) Sliding rod

Motion About a Fixed Point



A Rigid Body In Translation

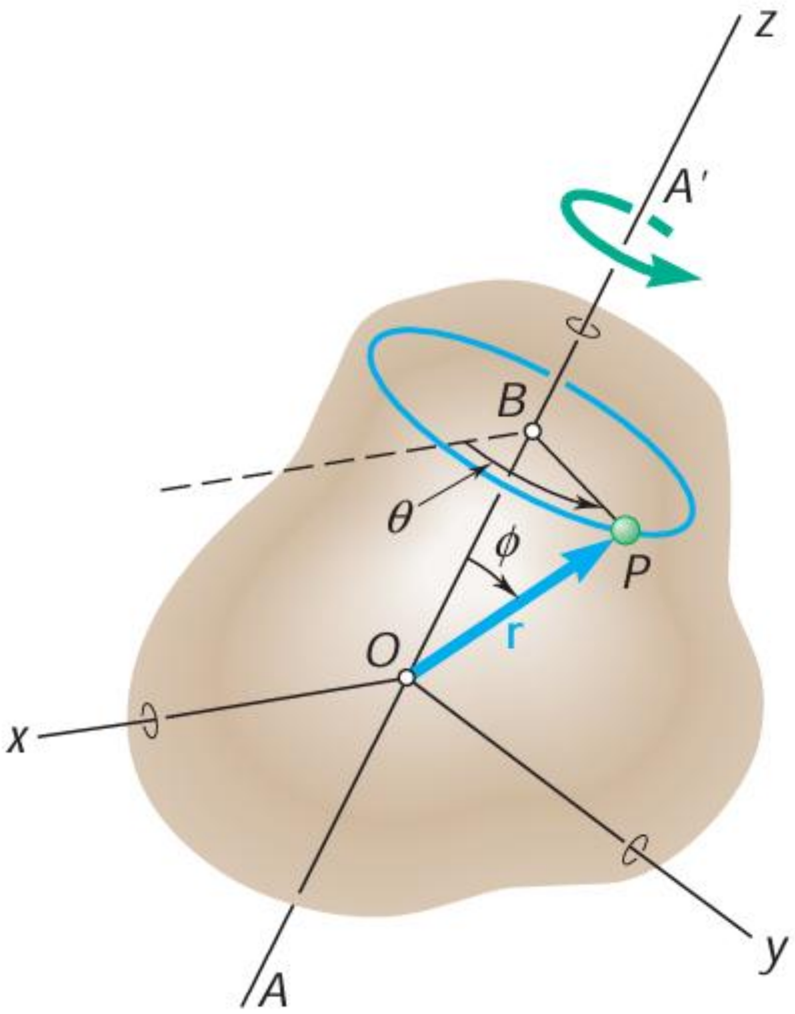
When a rigid body is in translation, all the points of the body have the same velocity and the same acceleration at any given instant



$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A}$$

$\boldsymbol{r}_{\mathbf{B/A}}$: constant direction & constant magnitude

Rotation About a Fixed Axis

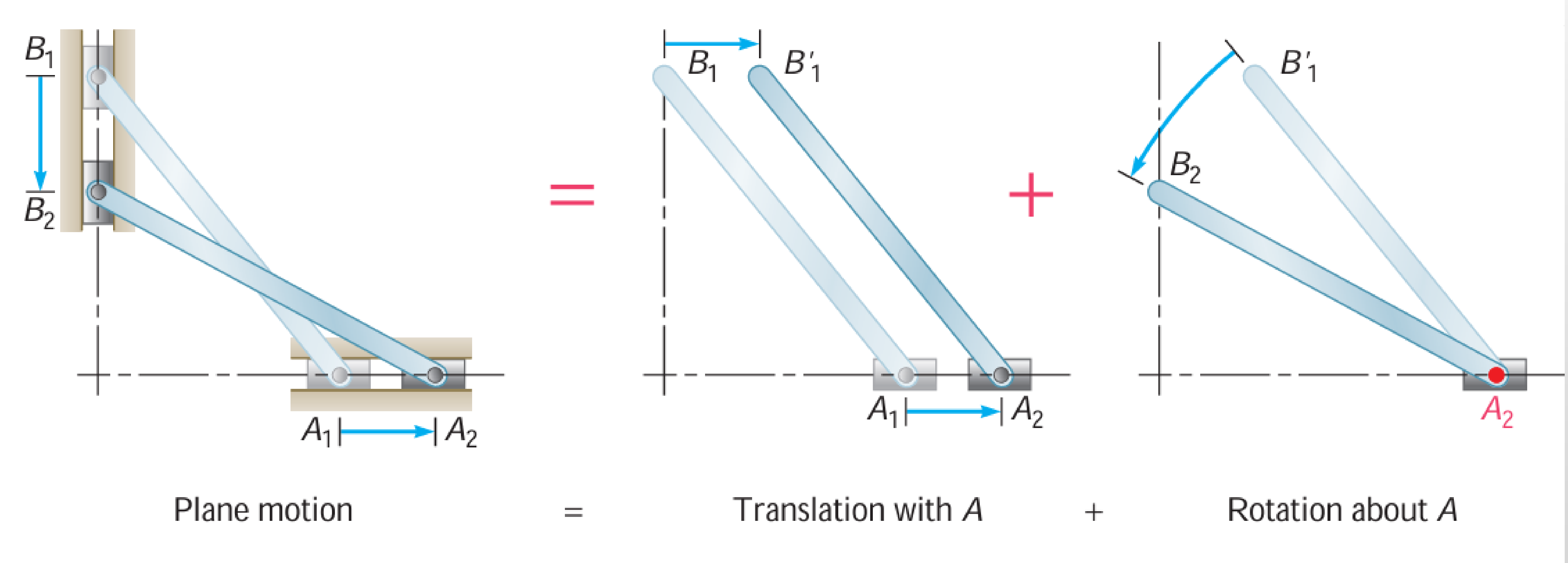


$$\Delta s = r\sin\varphi \times \Delta\theta$$

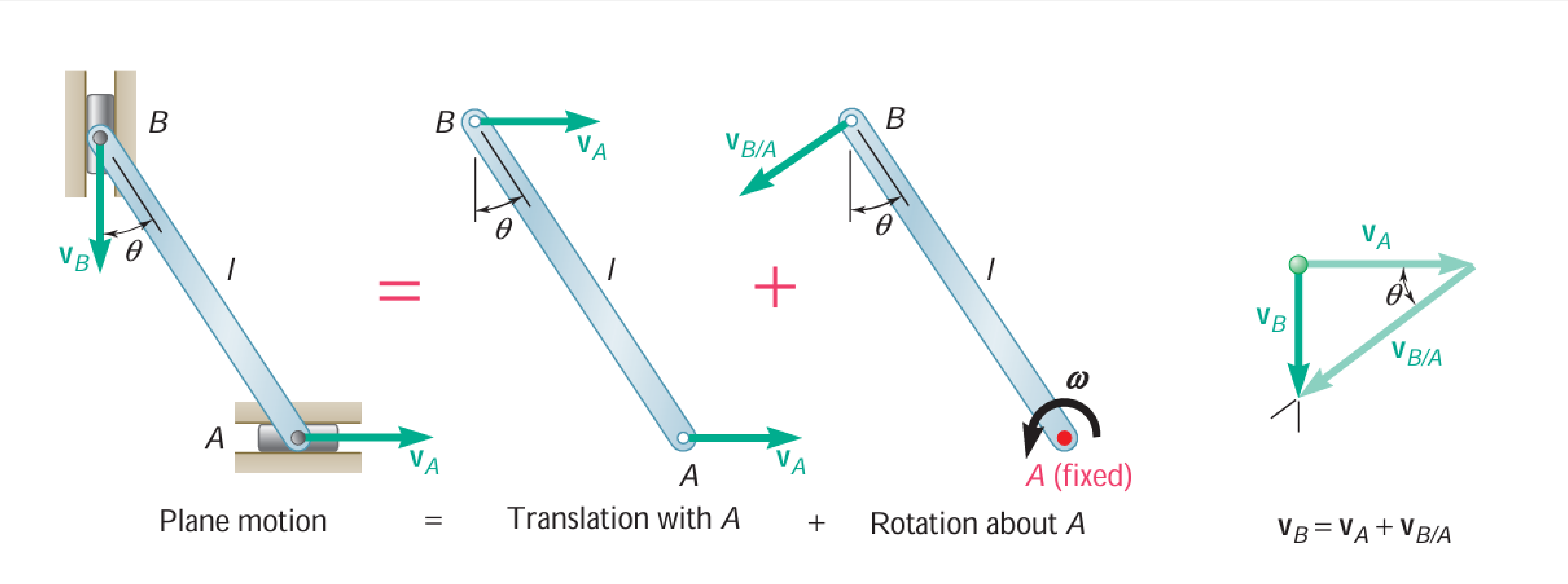
$$\lim_{\delta t\to 0}\frac{\Delta s}{\Delta t} = (r\sin\varphi)\lim_{\delta t\to 0}\frac{\Delta\theta}{\Delta t}$$

$$v = \omega \times r$$

General Plane Motion



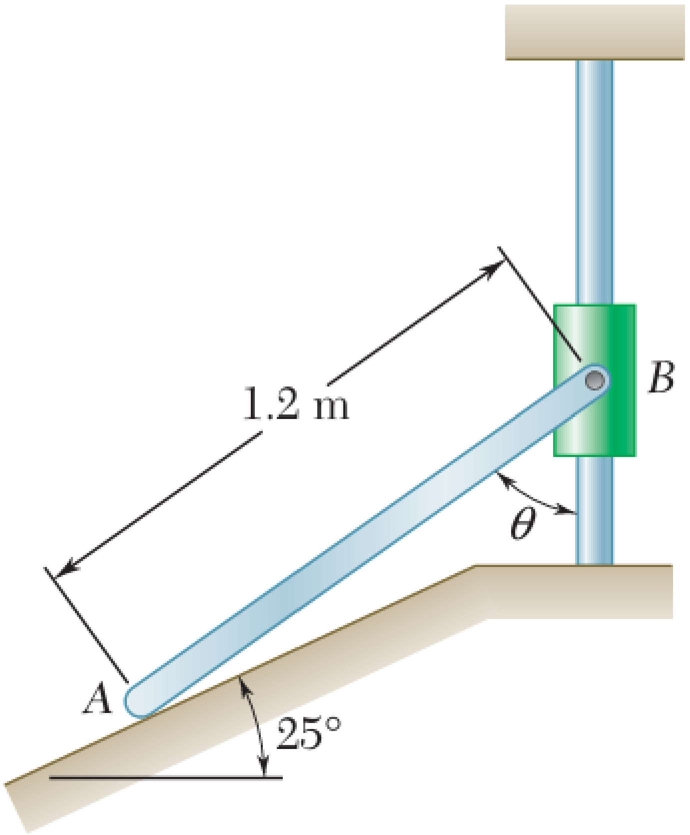
General Plane Motion



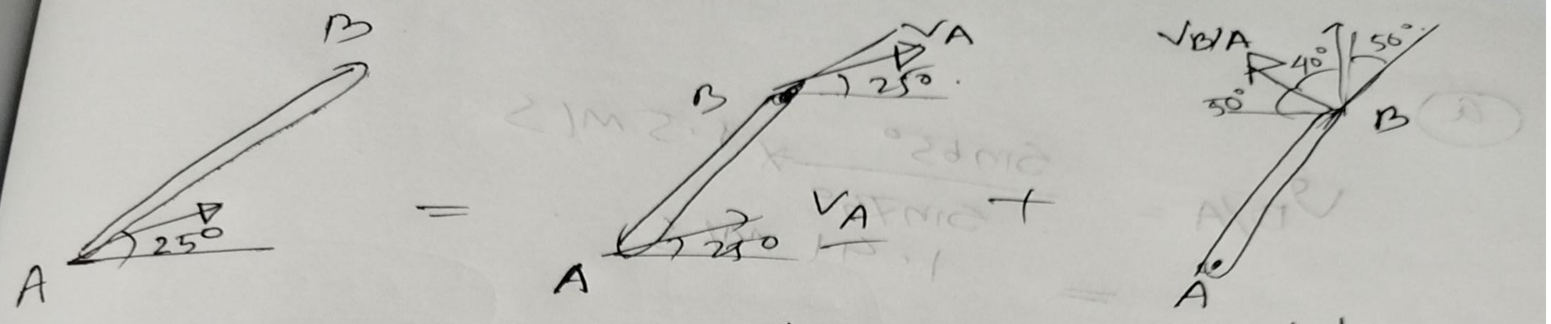
Kinematics

Problem:

Collar B moves upward with a constant velocity of 1.5 m/s. At the instant when $\Theta = 50°$, determine (a) the angular velocity of rod AB, (b) the velocity of end A of the rod.



Solution



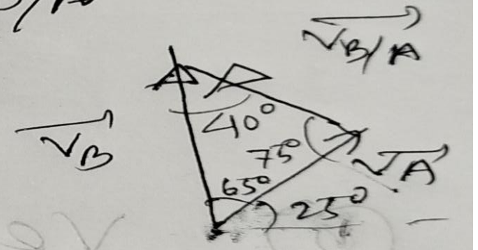
Plane Motion = Translation + Rotation.

$\vec{V_A} = V_A \angle 25°$

$\vec{V_B} = 1.5\ m/s \uparrow$

$\vec{V_{B/A}} = V_{B/A} \angle 50°$

$\vec{V_B} = \vec{V_A} + \vec{V_{B/A}}$



Sine Law of Triangle:

$\frac{V_B \rightarrow 1.5\ m/s}{\sin 25°} = \frac{V_A}{\sin 40°} = \frac{V_{B/A}}{\sin 65°}$

$1.5\ m/s \times \frac{\sin 65°}{\sin 75°} = 1.49\ m/$

(a) $V_{B/A} =$

$V_{B/A} = \omega_{AB}\, l_{AB} \Rightarrow \omega_{AB} = 1.49$

Solution:

(a)

$$v_{B/A} = \frac{\sin 65^\circ}{\sin 75^\circ} \times 1.5 \text{ m/s}$$

$$= 1.41 \text{ m/s}$$

$$v_{B/A} = \boxed{\omega_{AB}}\, l_{AB}$$

$$\omega_{AB} = \frac{1.41 \text{ m/s}}{1.2 \text{ m}} = 1.17 \text{ rad/s}$$

(b) Velocity of End A.

$$v_A = \frac{\sin 40^\circ}{\sin 75^\circ} \times 1.5 \text{ m/s}$$

$$= 0.998 \text{ m/s} \angle 25^\circ$$